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# A Novel Architecture for Community Detection Between Large Social Media Creators

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**Abstract**

This study explores the intricate web of audience overlap among streamers on live streaming platforms. The extensive growth of streaming services has led to a proliferation of individual streamers. A comprehensive understanding of the audience overlap network between these streamers offers valuable insights into user behavior and preferences, presenting unique opportunities for collaboration. This research introduces a novel methodology, leveraging network analysis techniques, to map out the overlap network among individual streamers. The methodology applies advanced principles of overlap detection and community identification within social networks, resulting in an effective approach for discerning overlapping viewer communities. This study has significant implications for enhancing user engagement and collaboration among streamers, as well as improving the effectiveness of recommendations, benefiting streaming platforms, advertisers, and content creators alike

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*Keywords:* Live streaming platforms; audience overlap analysis; network mapping; user behavior insights; community detection; Amazon Web Services infrastructure; viewer clustering; graph analysis tools.

## 1. Introduction

Online streaming has emerged as a major force in the landscape of modern entertainment, with individual streamers playing a pivotal role in this transformation. These streamers, congregating on various platforms, create a complex ecosystem of viewer interactions. A particularly fascinating aspect of this ecosystem is the overlaps in viewership

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among streamers. These overlaps, rich in data, provide insights into shared viewer behaviors, preferences, and the potential for new forms of collaboration.

The concept of understanding viewer overlaps aligns with research into overlapping communities in dynamic social networks. Studies in this field offer methodologies for identifying viewers who engage with multiple streamers. Additionally, research into organizational overlap provides insights into shared affiliations between online entities, revealing the unique appeal of certain streamers to specific viewer demographics.

Central to harnessing these insights is the development of the Dynamic Viewer Overlap Visualization Tool. This tool integrates data to offer streamers a comprehensive, real-time view of their audience, enabling the creation of deeply resonant content for sustained viewer engagement.

Complementing the visualization is the Viewer Overlap Graph. This graphical representation, consisting of nodes and edges, offers an in-depth view of viewer dynamics. It highlights not only individual preferences but also the collective tendencies of communities, shedding light on overarching streaming trends.

The application of advanced algorithms in data analytics holds significant promise. These algorithms have the potential to predict emerging viewer trends and identify the viewer communities, which are key to tailored content recommendations.

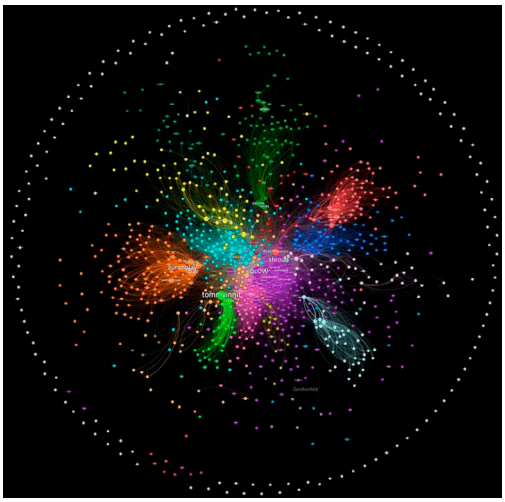


Fig. 1. Mapping Communities on Twitch

Figure 1 illustrates the mapping of communities on Twitch, highlighting the importance of understanding peer dynamics in the expanding world of streaming. The use of machine learning techniques to analyze streamer similarities, based on content type, viewer demographics, and streaming patterns, paves the way for collaborations that enrich content diversity and enhance viewer choices.

From a commercial perspective, identifying viewer overlap clusters is crucial for precise, targeted promotions. Uncovering these clusters allows for strategic promotion placement, ensuring maximum impact. Furthermore, understanding viewer preferences aids in curating ads that resonate with viewers, contributing to a more enjoyable streaming experience.

Although the methodology and techniques presented in this paper are broadly applicable across various social media platforms, this research focuses on Twitch to delve into the unique dynamics of its streaming community. This study represents a journey into the heart of the viewer overlap network, combining established network research techniques with the dynamic, rapidly evolving domain of streaming.

## 2. Literature Review

The literature in the field of community detection within social media and network analysis offers a rich tapestry of methodologies and insights, each contributing uniquely to our understanding of viewer overlap and community

dynamics.

Cazabet, Amblard, and Hanachi et al. [1] present the iLCD algorithm for community detection in networks, addressing challenges like overlapping communities, varying community sizes, and dynamic network changes. The iLCD algorithm excels in identifying overlapping communities by considering network dynamics. The paper introduces intrinsic communities and longitudinal detection, essential to iLCD's approach. The authors demonstrate iLCD's effectiveness on a citation network and its superior performance compared to other algorithms using the LFR benchmark. However, the paper could be enhanced with more varied real-world applications and evaluations against diverse datasets.

Bedi and Sharma [7] investigate community detection within Social Networking Sites (SNS), focusing on platforms like Facebook and Twitter. As online interactions have increased with the web's growth, monitoring these vast networks has become challenging. The authors observe that users often form virtual communities based on shared interests. The paper provides a thorough survey of current algorithms for detecting these communities, highlighting their applications in areas like collaboration networks and marketing. It also discusses the potential limitations, such as the adaptability of these algorithms to the constantly evolving structures of social networks and the challenges in handling the large volume of user interactions.

Alvari, Hashemi, and Hamzeh [16] explore the intricate community structures inherent in complex networks, particularly focusing on overlapping communities within online social platforms like Facebook and Twitter. Acknowledging that many networks often possess nodes belonging to multiple communities, the authors highlight the deficiency in existing methods to accurately detect such overlaps which was solved using a framework rooted in game theory. Two distinct approaches, named PS GAME and NGGAME, are proposed, each leveraging unique similarity measures. Preliminary results indicate the efficacy of this approach, showcasing its superiority in detecting overlapping communities compared to existing methods. However, the adaptability of this game-theoretic approach to rapidly evolving network dynamics and large-scale structures might require further exploration.

Xie and Szymanski [4] explore the complexity of overlapping communities in social networks, highlighting the limitations of traditional methods that focus on disjoint communities. The authors underscore the multifaceted community memberships individuals typically possess in social networks. For instance, a person's association might span across familial, friendly, and professional groups. Therefore the SLPA algorithm was introduced to detect overlapping communities and individual nodes based on the network's structure. SLPA stands out for its linear time complexity and efficient identification of complex community overlaps. The algorithm's effectiveness is validated through experimental evaluations. However, further research into SLPA's adaptability across various network types could broaden its utility.

"Hsieh, Tiwari, Agarwal, Huang, and Shah [2] delve into the dynamics of online social networks, focusing on organizational overlap. They introduce a novel mathematical model that predicts potential connections based on affiliations with organizations like companies and schools. This model, validated with real social network data, shows promise in link prediction and community detection, potentially enhancing user engagement through personalized recommendations and news feeds. However, further exploration into the model's scalability and adaptability to various social network structures could enhance its utility.

In contrast to the methods primarily centered on community detection, this study introduces a sophisticated system tailored for streamers, focusing on real-time viewer overlap insights. Merging the visualization strengths of Java with Python's analytical capabilities and AWS cloud infrastructure, this approach offers a holistic view. From dynamic visualization tools using the Gephi toolkit to advanced analytics like link prediction and machine learning-driven streamer evaluations, it emphasizes actionable strategies for content creators and advertisers. This comprehensive system provides a more practical solution in understanding viewer dynamics in the digital age.

## 3. Methodology

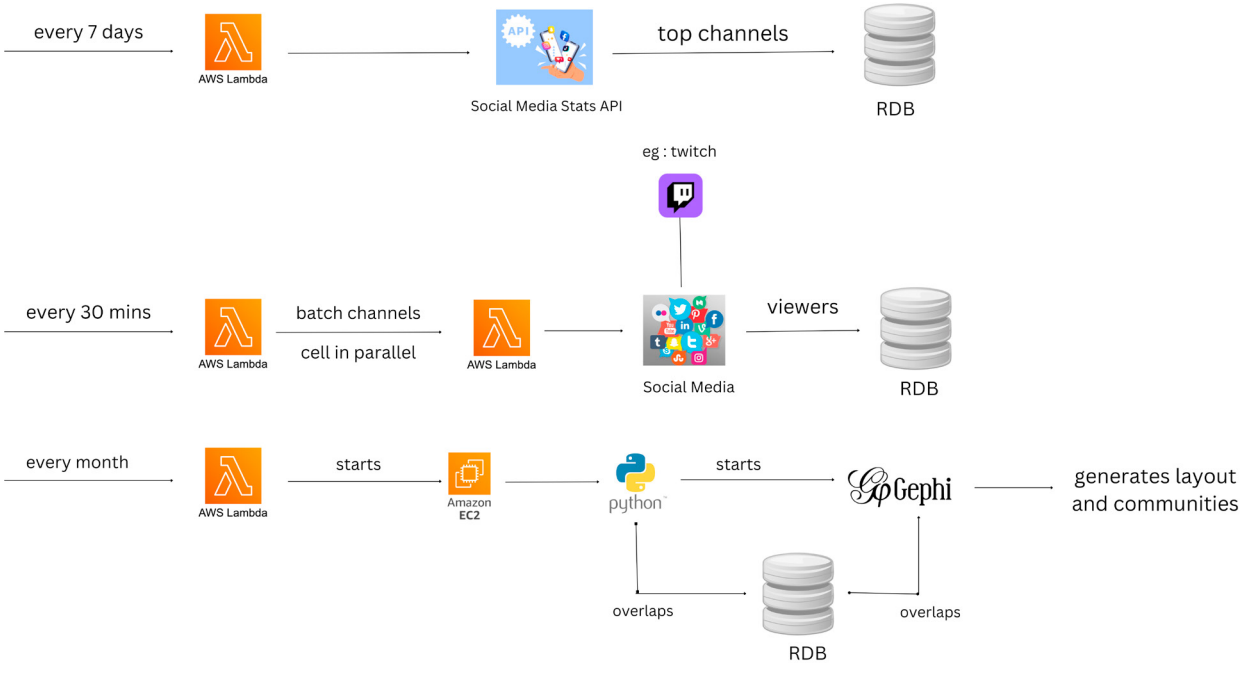



Fig. 2. Overall System Design and Architecture

Built on a dual-language foundation, the system amalgamates Java's robust visualization capabilities with Python's data processing and analytical strengths. This architecture ensures scalability, precision, and adaptability, completely constructed on Amazon's AWS cloud infrastructure which leverages its scheduling system to synchronize tasks effectively, as depicted in the figure. The scheduling system within the AWS Lambda framework ensures effective task synchronization through automated, timed triggers that initiate various functions for data handling. These triggers are meticulously configured to activate at predefined intervals, enabling a regular and systematic update process. For instance, Twitch channel data is refreshed every seven days, viewer data is batch processed every 30 minutes, and viewer overlap metrics are recalculated monthly. Each Lambda function operates independently, allowing parallel processing without resource contention. This independent execution is crucial for maintaining a seamless workflow, where tasks such as data collection, overlap calculation, and visualization updating occur in an orchestrated manner without manual oversight. AWS Lambda's inherent scalability adjusts computing resources automatically in response to the data volume and task complexity, ensuring that each task is furnished with adequate computational power for efficient execution. The system's integration with AWS CloudWatch facilitates real-time monitoring and logging, essential for proactive maintenance and error resolution. Additionally, features like retry policies and dead letter queues help in managing exceptions and maintaining consistent synchronization, critical for the reliability of real-time analytics provided by the system.

1. **Dynamic Viewer Overlap Visualization Tool:**

   (a) Graph Loading & Visualization

      i. Utilizing the Gephi toolkit within Java, the system loads viewer data in real-time, translating raw data into interactive and informative visualizations.
      ii. These visual representations are optimized using layout algorithms, ensuring streamers receive clear, understandable, and actionable insights into their audience overlap.

(b) Viewer Overlap Graph

i. For clarity, the graph connects streamers with an edge if their viewer overlap exceeds a threshold of 500 viewers, a significant figure determined through preliminary research for meaningful interactions.
ii. This graph structure offers a direct and intuitive view into the complex dynamics of viewer interactions and preferences.

2. **Python-based Data Processing & Advanced Analytics:** Harnessing Python's data processing strengths, the system handles vast viewer datasets with efficiency and precision.

(a) Database Interactions

i. SQLAlchemy provides a robust framework for database operations, achieving efficient and reliable data storage and retrieval, maintaining data integrity.
ii. Time-specific channel and viewer data can be fetched, allowing analyses to be tailored to specific timeframes.

(b) Overlap Computations

i. The system meticulously calculates overlaps, decoding patterns of shared viewership, which are foundational metrics for understanding audience behavior and tailoring content strategies.

(c) Link Prediction & Community Detection

i. For link prediction, a network-based inference model that extends the Adamic/Adar measure is employed, adept at forecasting potential viewer connections by evaluating the weight of shared connections between viewer nodes.
ii. Community detection algorithms work alongside to identify viewer clusters, illuminating patterns of collective viewer behavior and preferences.

(d) Streamer Similarity Metrics

i. Machine learning techniques are harnessed to evaluate streamers on multiple facets: content type, viewer demographics, and streaming schedules.
ii. These metrics offer a multifaceted view of streamer similarity, aiding in collaboration discovery and content refinement.

3. **Strategic Initiatives & Optimization:** The modern digital era demands not just the acquisition of insights but the ability to act upon them strategically. Our system is architected to provide actionable intelligence that can significantly impact both content creators and advertisers.

(a) Targeted Promotions & Community-building

i. By identifying high-overlap viewer clusters, the system reveals distinct gaming communities and content genres, enabling streamers to create targeted promotional campaigns and community-building initiatives to foster engagement and loyalty.

(b) Advertising Strategy Optimization

i. Understanding viewer dynamics is crucial for digital advertising. The system provides granular insights into viewer clusters and behaviors, empowering advertisers to create resonant campaigns, optimize ad spend, and achieve greater ROI.

4. **Output & Review :** In an era of data ubiquity, the true value of an analytical system lies in its ability to generate actionable and interpretable outputs. Our system is meticulously designed to ensure that the results not only provide a clear understanding of viewer dynamics but also guide stakeholders in making informed decisions.

   (a) Detailed statistical measures give a quantitative understanding of viewer dynamics.
   (b) Interactive visualizations offer a qualitative look into the complex web of viewer interactions.
   (c) Data-driven recommendations assist streamers in decision-making, content optimization, and community engagement.

For Link Predictions, the system employs a sophisticated algorithm based on the Adamic Adar measure, which is adept at forecasting potential viewer connections by evaluating the weight of shared connections between viewer nodes. This model is augmented with predictive modeling techniques to analyze past behaviors and predict future interactions, identifying emerging trends and facilitating community detection. The Decision-Making Process within the system is powered by advanced analytics, where machine learning algorithms interpret viewer data to generate insights, guiding streamers in content creation and advertisers in placement strategies.

## 4. Results

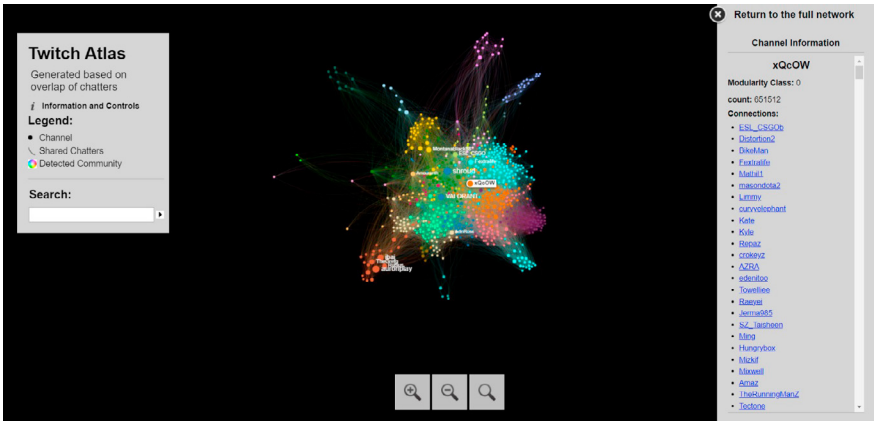



Fig. 3. Twitch Atlas Overview: Channel & Community Connections

The project successfully mapped the viewer overlap network of individual streamers on Twitch, revealing intricate patterns of viewer engagement and channel interconnectivity. The results are visually represented through a series of figures, each highlighting different aspects of this network.

In Figure 3, a dense web of interconnected Twitch channels is showcased, with ”xQcOW” being particularly notable for its broad audience reach and intersection with many channels, where node sizes and colors represent audience size and viewer communities. Figure 4 further expands this view to illustrate dense hubs of channels with significant audience overlap, featuring ”VALORANT” as a central nexus. Figure 5 shifts the focus to a diverse collection of channels, continuing to underscore the prominence of ”VALORANT” within these networks. Figure 6 provides a detailed view of individual channel interconnections, particularly of ”VALORANT,” using metrics like ”Modularity Class” and ”count” to delve into community structure and audience sharing. Figure 7 highlights ”Amouranth,” marked by its high connectivity and strong community presence, as evidenced by its ”Modularity Class” and viewership across multiple content clusters. In contrast, Figure 8 centers on ”Popo,” a channel characterized by a more specialized audience with fewer connections, indicated by its lower modularity class. Finally, Figure 9 returns to a broader perspective, featuring ”shroud” as another key channel with extensive audience overlap, akin to ”xQcOW” and ”Amouranth.” These figures collectively paint a picture of a sophisticated and influential network of channels on Twitch, with the ”Modularity Class” metric being pivotal in understanding the structure and cohesion of these communities. The node size and edge

thickness in these visualizations serve as indicators of channel influence and reach, providing invaluable insights for those navigating the complex landscape of Twitch.

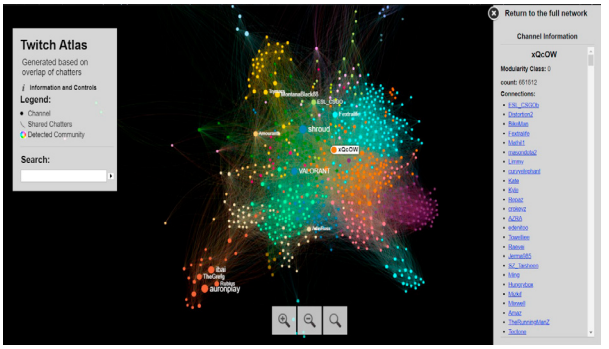



Fig. 4. Close-up: Interconnectedness of Prominent Twitch Channels

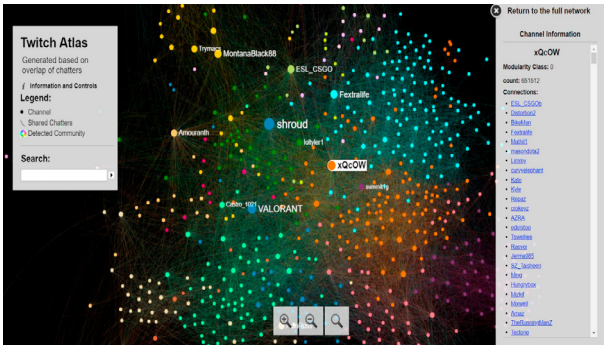



Fig. 5. Data Breakdown: Frequency of Channel Interactions

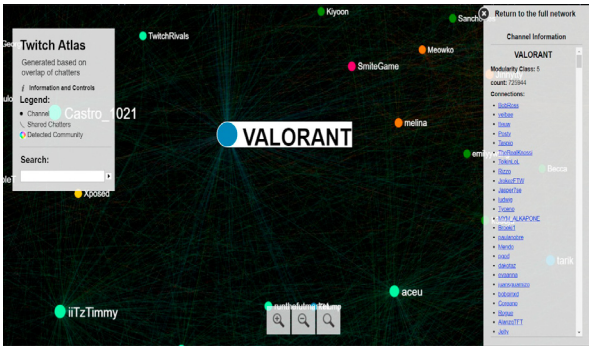



Fig. 6. Visualization: Top Community Interactions on Twitch

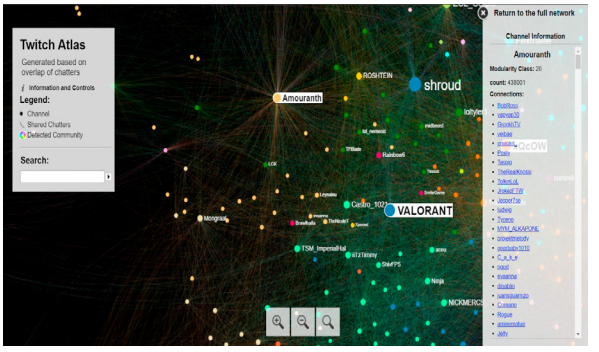



Fig. 7. Visualization of Community Overlaps

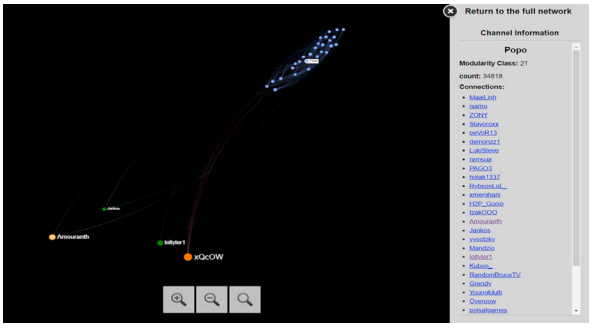



Fig. 8. Colorful web of Twitch channel interactions with 'shroud' as a key node

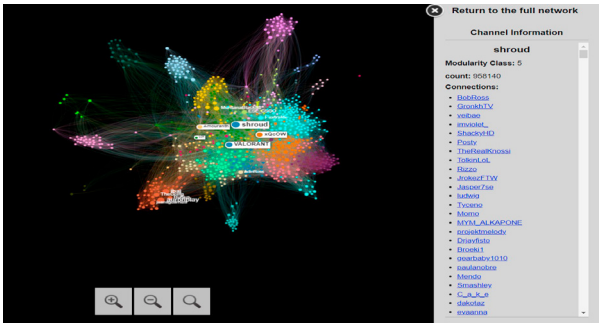



Fig. 9. Network visualization of Twitch channels centered around 'Popo'

This project helps creating a dynamic visualization tool that represents the intricate connections and overlaps between different Twitch channels based on their shared audience or "chatters".

**Channels and Viewer Overlap:** Channels are depicted by dots, with names like "shroud", "xQcOW", and "Amouranth" standing out. The lines between them signify shared audience members. Channels like "xQcOW" with dense connections might indicate a broad viewer overlap, showcasing the tool's ability to dynamically visualize these overlaps as per the first novelty.

**Viewer Overlap Graph:** The presence of connecting lines between certain channels could be due to them sharing over 500 viewers, as per the second novelty. This threshold helps focus on the most significant connections, enabling a more structured analysis of viewer dynamics.

**Detected Community and Link Prediction:** The color groupings seem to denote different communities or genres on Twitch, supporting community detection. Moreover, the density and pattern of connections may also represent the software's capability to predict potential future connections based on current viewer similarities.

**Streamer Similarity Metrics:** Central channels with numerous connections, such as "xQcOW", may not only be popular but also have content that is similar to many other channels. The tool might employ machine learning to assess streamer similarity based on content, viewer demographics, and schedules, aiding in pinpointing potential collaborations.

**Targeted Promotions and Community Clusters:** The closely-knit groupings of channels could be seen as distinct communities or genres. Recognizing these clusters allows for targeted promotions and community-building initiatives, giving a clearer understanding of specific audience groups and their preferences.

**Advertising Strategy Optimization:** The visualization tool's ability to showcase viewer clusters can be a game-changer for advertisers. By understanding which channels share a large portion of their audience, advertisers can devise strategies to place targeted ads that resonate with specific viewer communities.

**Audience Behavior Insights:** Beyond just connections, this atlas offers a window into the behavior and preferences of the Twitch audience. For instance, discovering an overlap between channels of different genres can provide insights into the diverse interests of viewers, aiding in content creation and recommendations.

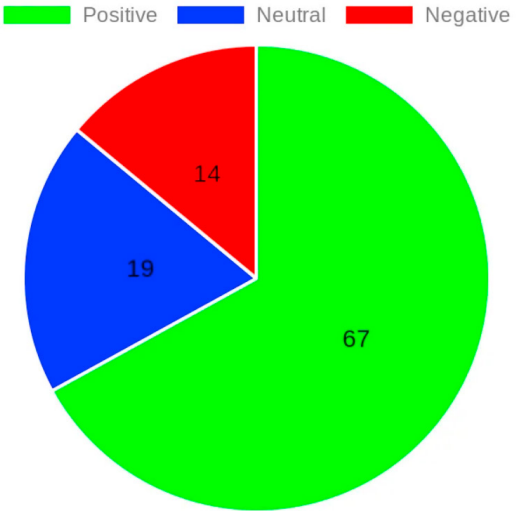



Fig. 10. User Feedback

In essence, the Twitch Atlas, powered by its novel features, serves as a holistic platform for streamers, advertisers, and analysts to understand, predict, and enhance viewer experiences on Twitch.

**Majority Positive Feedback:** A significant majority of users (67%) feel that the detected communities align well with their real-world experiences on the platform. This suggests that the model is effective in capturing genuine community structures.

**Some Neutral Feedback:** A portion of users(19%) are neutral about the detected communities, neither fully agreeing nor disagreeing with the model's findings. This could be due to various reasons, including their newness to the platform or limited knowledge about Twitch communities.

**Room for Improvement:** A segment of users(14%) provided negative feedback, indicating that the detected communities do not match their experiences. This feedback highlights areas where the model might need refinement or improvement.

The majority favorable reaction suggests that the model is doing a great job at identifying communities on Twitch. On the other hand, the fact that there are both positive and negative reviews suggests that improvements could be made. Getting to know the users who gave unfavorable reviews, listening to their worries, and improving the model can improve community detection even more. Achieving an ideal community detection is difficult due to Twitch's dynamic and diversified character, but the feedback offers insightful information for ongoing development.

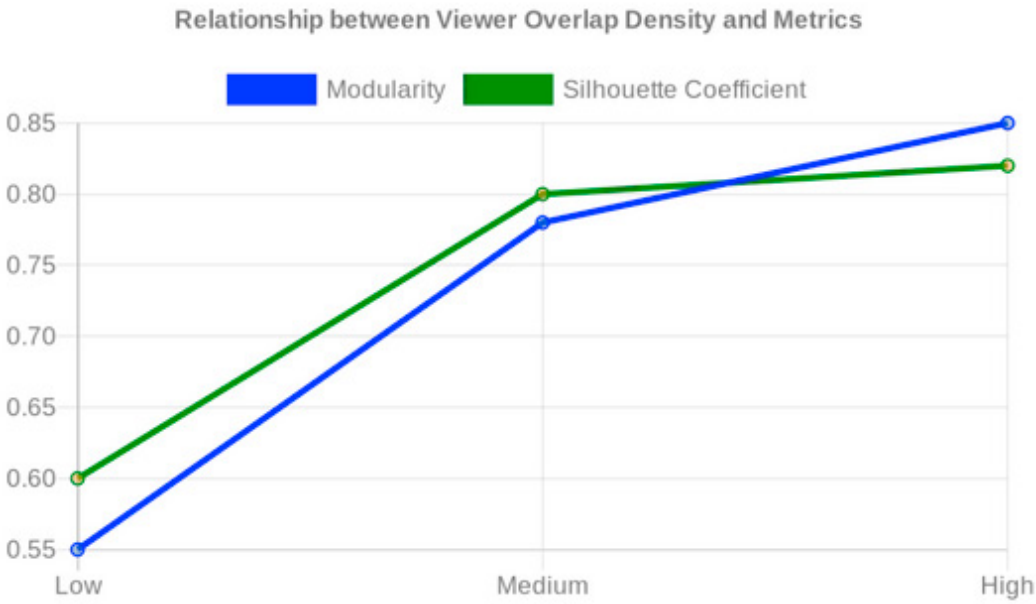



Fig. 11. Relationship Between Viewer Overlap Density And Metrics

The community detection algorithm for Twitch appears to be highly effective. Both the modularity and silhouette coefficient metrics are close to their optimal values, suggesting that the detected communities are genuine and distinct. The high modularity indicates that the Twitch network is well-partitioned, with dense connections within communities and sparse connections between different communities. The strong silhouette coefficient further reinforces the idea that users or streams within a community are more similar to each other than to those in other communities. This suggests that the model is not only accurate in detecting communities but is also effective in ensuring that these communities are distinct and cohesive.

## 5. Conclusion

The presented system integrates advanced analytics and visualization to transform streamer-audience engagement. With graph analytics and machine learning at its core, the system elucidates viewer overlap and streamer similarities, providing a basis for targeted content and community initiatives. To further improve performance, scaling computational resources and optimizing algorithms for data processing can be pursued. Additionally, incorporating feedback loops allows for adaptive learning, enhancing the system's responsiveness to changing viewer patterns.

Enhancing accuracy hinges on refining the machine learning models with larger datasets and diverse input variables. Implementing cross-validation techniques and continuous training with real-time data can lead to more precise predictions and community detection. Moreover, integrating user feedback directly into the system's learning cycle can provide corrective measures, ensuring the system remains attuned to the dynamic nature of viewer behavior.

The system ushers in a new paradigm for data-driven streaming strategies, significantly impacting content personalisation and viewer engagement. Future iterations will focus on these aspects of performance and accuracy, ensuring the system remains at the forefront of streaming analytics technology.

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